

Transparency and Manipulation of Public Accounts¹

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Abstract

We study the policy choices of an incumbent government when voters imperfectly observe aggregate spending. Total spending is decreasing in the transparency of spending but increasing in the transparency of the incumbent's ability.

The model provides a possible explanation of the choice of inefficient tools of redistribution, and investigates the incentives for politicians to manipulate public accounts. We show that politicians may choose inefficient manipulable tools. However, the existence of such tools may have positive welfare effects because it leads to a reduction in wasteful spending.

1 Introduction

This paper considers a model of an incumbent politician choosing transfers in a world of incomplete transparency of spending. We discuss the politician's incentives to manipulate public accounts and its consequences when such manipulation is associated with wasteful spending.

The issue of the transparency of government activity has recently received a lot of public attention. The common perception is that most spending is performed without sufficient monitoring. For instance, "more than 90 percent of earmarks are specified [just in committee reports], not actually included in the texts [of legislation]." ¹ In a recent article in the *New York Times*, ² we read: "Mr. Hollings [U.S. Senator], who contends that the government hides its debt no less egregiously than Enron did, asked Mitchell E. Daniels Jr., the White House budget director, to swear personally to the accuracy of the government's financial reports. [...] By any standard, Washington does not do a good job of keeping its own books. Mr. Daniels issued a report card this year on government performance that gave poor grades for financial management to every major cabinet department, and his own agency to boot. 'This is a serious subject,' Mr. Daniels said. 'No one could attest to the accuracy of the government's books.'" ³

This lack of transparency is often associated with abuse of pork-barrel spending and "earmarks", and this has spurred a call for reform. A general theme of most reform proposals is that increasing transparency is essential for maintaining fiscal discipline. For example, a central pillar of Obama's campaign platform was to "Shine the light on federal contracts, tax breaks and earmarks." The Obama campaign claimed that "Too often bills are rushed through Congress and to the president before the public has the opportunity to review them. As president, Obama will not sign any non-emergency bill without giving the American public an opportunity to review and comment on the White House website for five days." ^{4,5}

¹"Earmarks Likely to Continue, but With Details," *New York Times*, January 22, 2008.

²August 15, 2002.

³Similarly, in a recent opinion piece in the *Wall Street Journal* (April 18, 2006), Senator Judd Gregg laments that the federal government "is keeping two sets of books....the nation's elected leaders are abusing the process....The defense budget is a good example. One set of books said the Defense Department and related national security functions would require \$434 billion this year, about a 3% increase over the previous year. When emergency funding is factored into the equation, the true set of books show that defense will be provided at least \$558 billion, or almost a 12% increase above the previous year."

⁴<http://www.barackobama.com/issues/ethics/#earmarks>

⁵Transparency has also been an important focus for international organizations. For instance, the IMF published a book in 2001 titled *Manual on Fiscal Transparency* containing recommendations on how budgetary institutions and national accounts should be organized. The OECD published a similar volume in 2000 titled

In this paper, we develop a simple model that adapts standard models of career concerns to a political economy context in order to capture some key features of incumbent politicians' incentives in a world with imperfect transparency.⁶ The basic ingredients of the model are the following. An incumbent politician chooses transfers to maximize a weighted average of re-election probability and social welfare. Voters are *ex ante* identical, they are uncertain about the incumbent's ability, and they observe an imperfect signal of aggregate spending. Since taxation is distortionary, efficiency involves zero spending on transfers. However, this efficient outcome is not an equilibrium: the incumbent has an incentive to "cheat" and provide positive transfers because this would increase voter's beliefs about the incumbent's ability. Thus, in equilibrium all voters are offered positive transfers. We then show that the equilibrium level of transfers is higher when voters have more precise information about the incumbent's ability, and offer a possible interpretation of this result as one negative effect of political accountability. In our model, an increase in the precision of voters' information about the incumbent's ability is equivalent to an increase in the sensitivity of the incumbent's probability of re-election to his current actions. In our model, increasing this sensitivity leads to a more inefficient outcome. In prior work, Persson and Tabellini (2000) came to the opposite conclusion on the value of such accountability. This is because in their model the politician's choice is the amount of corruption, and increasing accountability reduces corruption.

We next consider the possibility that the incumbent government may choose wasteful means of transferring resources, *even* if non wasteful means are available.⁷ We first describe the welfare consequences of alternative transfer tools when the wastefulness of the transfer tools is *exogenously* given. We compare two regimes: in the first regime the politician can only use the wasteful transfer tool; in the second regime the politician can only use the non-wasteful tool. We show that, paradoxically, welfare is higher in the wasteful regime. The reason is that wastefulness endogenously constrains the size of government, leading to lower overall distortions by reducing the size of distortionary taxes. We then allow the incumbent to choose the type of transfer. The fact that welfare is higher in the wasteful regime induces (perhaps counterintuitive) choices by the incumbent: we show that if the incumbent can commit to the type of transfer, he chooses the wasteful tool; whereas if he cannot commit he chooses the non-wasteful tool. We then assume that the incumbent can manipulate public accounts by choosing different transfer tools. For instance, direct lump sum transfers are not wasteful, but relatively transparent. In contrast, distorting the construction of public projects

Best Practices for Budget Transparency.

⁶See Holmström (1999).

⁷This extension of the model is related to Coate and Morris (1995). We discuss this paper and the difference with our results in Section 3.4.

in a way that leads to rents for some groups is a less efficient way to transfer resources to such groups, but it is harder for voters to figure out the true cost of these transfers. In this context, we show that the incumbent may then favor wasteful transfers even without commitment if they are more manipulable.

This paper contributes to the large literature on the link between fiscal institutions and fiscal performance.⁸ Many contributions to this literature do not provide formal models. There is however a growing formal and empirical literature that examines the link between constitutions and economics policy. See Persson and Tabellini (2003) for a survey of this literature. This paper also contributes to the literature on the political economy of budget deficits. Persson and Tabellini (2000) provide a detailed survey of this literature. Recently, a small strand of the literature has connected economic policy with the transparency of the political process. Our paper is particularly close to work by Milesi-Ferretti (2003), Shi and Svensson (2002) and Alt and Lassen (2006). All these papers discuss models with incumbent politicians and obtain that higher transparency reduces incentives to accumulate debt. In prior work (Gavazza and Lizzeri 2009) we considered related questions in a model of electoral campaigns (as opposed to incumbent government) in which each voter observes an imperfect signal of offers made to other voters. As in this paper, we showed that, imperfect transparency of transfers leads to inefficiency so that increasing transparency of spending is beneficial. In contrast, we showed that transparency of revenues may be counterproductive. Mattozzi and Merlo (2007 a and b) provide models in which political parties compete with the private sector to hire talented workers. They find that an increase in the transparency of politics reduces the average quality of the politicians.

2 Model

There is a unit measure of voters divided in n equal sized groups.⁹ There is a single consumption good. Voters have preferences specified by a utility function given by

$$u = \theta - \psi(t),$$

where we interpret θ as a public good (or transfer) received by the voters, and $\psi(t)$ as the disutility cost associated with taxation t . We assume that $\psi(t)$ is increasing, convex, and

⁸See for instance, Von Hagen (1992), von Hagen and Harden (1994), and Poterba and von Hagen (1999).

⁹Most of our results would be essentially unchanged if we considered a representative voter. However, allowing for multiple groups presents no complication, so we chose to present the model in this more general form.

twice differentiable, and, in order to guarantee that taxation leads to distortions, we assume that $\psi'(0) = 1$.

The government chooses the amount of spending y and revenues t . The level of spending y together with the government's ability a in delivering useful public spending determine the benefits received by the voters in the following manner:

$$\theta = y + a. \tag{1}$$

We assume that a is distributed according to the density f . We assume that f is symmetric and unimodal. As in Persson and Tabellini (2000, chapter 4) and Alt and Lassen (2006), we assume that neither the incumbent nor the voters observe the ability a at the time when they make decisions.¹⁰ Voters make inferences about the abilities of the incumbent by observing realized values of θ and any additional signals on spending, as specified later.

There are three periods. In the first period, the government chooses policies $(y_1, \dots, y_n)$, $(t_1, \dots, t_n)$. For each group i , a policy (y_i, t_i) specifies spending y_i on a public good that benefits group i and revenues t_i raised from group i . Voters in group i observe θ_i and ψ_i , and possibly, signals of aggregate revenues (this will be made precise below). In the second period, voters choose to either re-elect the incumbent or elect the challenger. The challenger's abilities are drawn from the same distributions as the incumbent.¹¹ In the third period, the second period incumbent chooses optimal policies. For notational simplicity, we assume that there is no discounting.

The incumbent chooses policies in order to maximize a weighted average of probability of re-election and social welfare. Denote by α the weight that the incumbent places on the value of winning.¹²

2.1 Comments on modeling assumptions

There are two reasons for assuming that the incumbent's ability affects voters' utilities. First, there is evidence that the characteristics of individual leaders can have a significant effect (See, for instance, Besley and Case, 1995; Jones and Olken, 2005). Second, as in most models of incumbent behavior, voters condition their decisions on fiscal performance *because* of the effects that this performance has on their beliefs about the incumbent's ability. This

¹⁰The fact that the incumbent and the voters have symmetric ignorance about the incumbent's abilities is in the tradition of models of career concerns à la Holmström (1999).

¹¹Symmetry is the natural benchmark but we will comment in Section 3.5 on what happens if the incumbent and challengers are drawn from different distributions.

¹²Even if politicians were completely unconcerned with social welfare, they may still act as if they place some weight on social welfare because of a variety of institutional constraints.

conditioning generates the incentives for the incumbent to manipulate fiscal variable in order to increase their re-election probabilities.

We have made several simplifying assumptions. First, we have formulated a reduced-form payoff function over policies (e.g. disutility of taxation) rather than deriving these preferences from more basic primitives, such as taxation over labor supply. This is mainly for convenience and because of the focus of our analysis: in this paper we focus on broad inefficiencies introduced by imperfect information about policies instead of delving into the details of the economic activities that are affected by these distortions. Second, we have made stark functional form assumptions: linearity in transfers, separability of benefits of spending and costs of taxation, and additive ability. These assumptions permit a simple characterization of equilibrium policies, and this is helpful for interpreting the effects of transparency. We could accommodate alternative interactions between ability and transfers; for instance, a multiplicative form would lead to similar results. Similarly, we could allow for concave utility functions, at the cost of more burdensome formulations.

3 Analysis

We first obtain an expression for social welfare over the two periods. Denote aggregate spending by $Y = \sum_{i=1}^n y_i$, and aggregate revenues by $T = \sum_{i=1}^n t_i$. Because in the last period there are no electoral concerns, the government will just maximize second period social welfare. Thus, given any deficit cumulated in the first period, the optimal choice in period two is to repay the debt with all voters bearing an equal share (because of convexity in distortionary taxation). Recalling that aggregate debt is given by $D = Y - T$, welfare is thus given by the sum of utilities over the two periods. Because ability enters additively, we can ignore it and write the expression for welfare as follows:¹³

$$SW = \frac{1}{n} \sum_{i=1}^n y_i - \frac{1}{n} \sum_{i=1}^n \psi(t_i) - \psi\left(\frac{Y - T}{n}\right).$$

3.1 Benchmark: Perfect Transparency

There are two cases to consider. First, we assume that there is perfect information about the incumbent's ability a (or degenerate distribution of ability). In this case, there is no electoral motive to provide positive transfers. Because of distortionary taxes, social welfare is

¹³The following expression assumes that there is never a surplus. It is easy to see that this is the relevant case.

decreasing in transfers. Thus, in the case of observable ability, the best that the government can do is to choose zero transfers and taxes.

Second, if voters perfectly observe transfers, the incumbent cannot fool voters by manipulating fiscal policy because ability can be deduced perfectly by subtracting fiscal policy measures from payoff outcomes: $a = \theta - y$. This implies that the probability of re-election is independent of policy choices, and, once again, the best the government can do is to choose zero transfers and taxes.

3.2 Imperfect Transparency

From now on, we assume that voters do not perfectly observe the government's policy choice, and that a is unknown to both voters and the incumbent. Voters receive a signal on aggregate spending: with probability p voters observe aggregate spending Y perfectly, with probability $1 - p$ each group i only observes θ_i .

Given this information structure, there are two possible information states. In each of these states, we need to obtain how the re-election probability depends on the incumbent's policies. We use the notation $\hat{y}_i$, $\hat{\theta}_i$, and $\hat{a}$ to denote voters' conjectures about the variable of interest.

- i. With probability $1 - p$ voters have no information about aggregate policies. We show that, in this event, the probability that group i votes for the incumbent is equal to $1 - F(\hat{y}_i - y_i)$. We first derive beliefs about ability for a generic group of voters conditional on any given observed θ_i . Clearly, the following consistency conditions between observed levels, conjectured choices and conjectured abilities must hold:

$$\theta_i = \hat{a} + \hat{y}_i \tag{2}$$

We can therefore obtain that the value of conjectured ability $\hat{a}_i$ for any θ_i is $\hat{a}_i = \theta_i - \hat{y}_i$. Substituting from equation (1) we obtain: $\hat{a}_i = a + y_i - \hat{y}_i$. In order to obtain the probability that the incumbent attributes to group i voting for him, we use the fact that the optimal choice for a voter is to vote for the incumbent as long as he believes that next period welfare under the incumbent is higher than under the challenger. Because we assumed that the ability of the challenger has the same expectation as the incumbent's, the voter votes for the incumbent if and only if $\hat{a}_i > E(a) \equiv 0$. Thus, evaluated at the moment in which the incumbent chooses policies (recall that the incumbent does not know his ability at that point), the probability that a voter in group i votes for the

incumbent given y_i is

$$\begin{aligned}\Pr(\hat{a}_i > E(a)) &= \Pr(\hat{a}_i > 0) = \Pr(a + y_i - \hat{y}_i > 0) \\ &= 1 - F(\hat{y}_i - y_i)\end{aligned}\tag{3}$$

- ii. With probability p voters observe aggregate transfers Y . We now show that, in this event, the probability that group i votes for the incumbent is equal to $\frac{1}{2}$. Consider a policy vector $(y_1, \dots, y_n)$ and associated conjectures $(\hat{y}_1, \dots, \hat{y}_n)$. Consistency of conjectures requires that

$$\theta_i + \sum_{j \neq i} \hat{\theta}_j = Y + n\hat{a};$$

and that

$$\hat{a}_i = \frac{\theta_i + \sum_{j \neq i} \hat{\theta}_j - Y}{n} = \frac{na + \sum_{i=1}^n \hat{y}_i - Y}{n}$$

Thus,

$$a = \hat{a}_i + \frac{Y - \sum_{i=1}^n \hat{y}_i}{n}.\tag{4}$$

Since voters observe Y , consistency requires that

$$\sum_{i=1}^n \hat{y}_i = Y.$$

We can conclude that equation (4) implies that, when voters observe Y , a is revealed to all voters before the election. Thus, at the time the incumbent chooses transfers, he needs to take the expectation over his ability a so the incumbent's probability of being re-elected is $\frac{1}{2}$ independent of the chosen policy.

We now need to aggregate individual voting decisions to obtain the probability of victory. Since all voters in a group vote in the same manner, in order to get re-elected the incumbent needs to win at least 50 percent of the groups.

The next Lemma obtains a convenient representation of the incumbent's re-election probability that we later use to derive the equilibrium.

Lemma 3.1 *Consider an equilibrium strategy profile $(y_1^*, \dots, y_n^*)$ and consider a deviation (y_i, y_{-i}^*) . The probability of re-election $\Lambda(y_i)$ can be expressed as*

$$k + \rho(n)(1-p)(1 - F(y_i^* - y_i))$$

where k is a constant that does not depend on y_i and $\rho(n) \equiv \left(\frac{1}{2}\right)^{n-1} \left(\frac{n-1}{2}\right)$.

Proof. See the Appendix. ■

Lemma 3.1 highlights how voters' imperfect information about government's policies affects the incumbent's incentives to choose transfers and taxes in order to influence voters' perception of his abilities. For example, the equation for $\Lambda(y_i)$ confirms that, if $p = 1$, the expected re-election probability does not depend on the incumbent's policies, since voters perfectly infer his abilities.

A profile of transfers $(y_1^*, \dots, y_n^*)$ constitutes an equilibrium if and only if, for every i , y_i^* maximizes the incumbent's objective function

$$\alpha \Lambda(y_i) + (1 - \alpha) SW.$$

The next Proposition characterizes the incumbents' choice of transfers and taxes.

Proposition 3.2 *In equilibrium:*

(i) *Aggregate first period transfers Y , and individual taxes t are characterized by the following conditions*

$$\psi' \left(\frac{Y - T}{n} \right) = \psi'(t) = 1 + n\rho(n) \frac{\alpha(1-p)}{(1-\alpha)} f(0); \quad (5)$$

(ii) *For any $p < 1$, debt and first period transfers are positive.*

Proof. The first order condition for the incumbent is:

$$\alpha\rho(n)(1-p)f(y_i^* - y_i) = -(1-\alpha) \frac{1}{n} \left(1 - \psi' \left(\frac{Y - T}{n} \right) \right).$$

Since $y_i = y_i^*$, we can simplify the above equations to obtain the following equilibrium condition:

$$n\rho(n)\alpha(1-p)f(0) = -(1-\alpha) \left(1 - \psi' \left(\frac{Y - T}{n} \right) \right). \quad (6)$$

Because the incumbent's first period choice of taxation does not affect his re-election probability, he chooses $t_1, \dots, t_n$ to maximize social welfare *given* the equilibrium choice level of aggregate transfers Y . Thus, he chooses $t_1 = \dots = t_n \equiv t$, and t such that $t = \frac{Y-T}{N}$ in order to smooth distortions across periods. Thus, rearranging equation (6), we obtain equation (5).

The fact that debt and first period transfers are positive for $p < 1$ is an immediate consequence of equation (5). ■

Equation (5) captures the conflict between the incumbent's desire to increase his re-election probability and his wish to limit the harm to voters' welfare. Equation (5) determines the equilibrium aggregate transfers Y . Because utility is linear in the transfer y , our model does

not pin down how equilibrium aggregate spending Y is divided into transfers to individual groups.¹⁴ Equation (5) equates the marginal second period distortions due an increase in first period transfers (and hence an increase in per capita debt $\frac{Y-T}{n}$) to a weighted sum of voters' marginal benefit from an additional transfer (equal to 1) and the incumbent's marginal increase in his re-election probability.

The intuition for part (ii) of Proposition 3.2 is the following. Transfers have to be positive in equilibrium otherwise the incumbent could increase his re-election prospect with minimal harm to social welfare. In particular, if voters expect zero transfers in equilibrium, the incumbent government could deviate by offering a small amount of positive transfers. This deviation only has a second order effect on welfare because $\psi'(0) = 1$, but it has a first order effect on the re-election probability. When—with probability $1 - p$ —voters do not detect the deviation, they are more likely to vote for the incumbent because their beliefs about his ability increases: they attribute their higher first period utility to ability rather than to increased spending. This effect captures one key element of our model: when voters do not perfectly observe transfers, the incumbent is held only partially accountable for the future distortions from current transfers. Therefore, the incumbent government do not fully internalize such losses. Hence, in equilibrium, transfers must increase up to the point where the marginal distortion from future taxes offsets the current electoral gain.

3.3 The Effects of Transparency

We now investigate the effects of two dimensions of voters' information on equilibrium fiscal policy. First, we consider the effect of changing the precision of voters' information about aggregate transfers and revenues. Second, we describe the effects of changing the precision of voters' information about the incumbent's ability, namely, the transparency of information about the quality of politicians. We model this dimension of transparency as an increase in the density $f(0)$. Recall that we are assuming that f is symmetric and unimodal. For all parametric classes of distributions with these characteristics, an increase in the mode corresponds to a reduction in the variance.¹⁵

Proposition 3.3 *The equilibrium has the following comparative statics:*

¹⁴If we extended the model to allow for a utility function concave in transfers, this indeterminacy would disappear.

¹⁵For instance, Normal, Laplace, Logistic, Student, Triangula, Error, Uniform. Note that, although the Uniform distribution is not unimodal, our results can accommodate the uniform distribution. See Evans et al. (1993).

- (i) *An increase in the transparency of transfers p leads to an decrease in transfers and debt, and an increase in social welfare;*
- (ii) *An increase in the transparency of incumbent's ability leads to an increase in first period transfers and debt, and a reduction in social welfare.*

Proof. To prove part (i), observe that the right-hand side of equation (5) decreases when p increases. Hence, debt, transfers and first period taxes decrease when p increases. Moreover, since transfers are wasteful, social welfare is increasing in p .

To prove part (ii), note that the right-hand side of equation (5) increases when $f(0)$ increases. Hence, debt, transfers and first period taxes increase since the function $\psi(\cdot)$ is convex. ■

Part (i) describes the comparative statics with respect to the transparency parameter p . The intuition follows closely from the intuition developed for the results of Proposition 3.2: when the transparency of transfers p decreases, the incumbent is more likely to offer them. Since transfers are socially wasteful, social welfare decreases as p decreases. Milesi-Ferretti (2004), Alt and Lassen (2006), and Gavazza and Lizzeri (2009) obtain a similar result.¹⁶ Part (i) of Proposition 3.3 also provides some justification for the importance of bodies such as the Congressional Budget Office and the General Accounting Office. Countries differ in the transparency of aggregate spending, partly because of the different role played by independent institutions that verify government accounts. Our model offers predictions about some of the effects of this heterogeneity.

Part (ii) of Proposition 3.3 reinterprets a classic comparative statics result in the career concerns literature (Holmström, 1999) within a political-economy context in which politicians may offer wasteful transfers.¹⁷ It says that transparency of ability may be harmful because the incumbent chooses a higher level of transfers and a higher level of debt. The intuition for this result is the following. Consider the effect of a marginal increase in transfers at the

¹⁶Our model is particularly close to Alt and Lassen (2006). However, they focus on debt, and on an empirical examination of debt across countries. Shi and Svensson, Alt and Lassen, and Alesina et al. (1999) also provide an empirical analysis of transparency. They look at cross-sections of countries and show that indices of transparency help predict fiscal outcomes: lower deficits are associated with better transparency. Prat (2005) studies a career concerns model where there is a distinction between information on the outcomes related to an agent's action and information directly on the agent's action. He shows that transparency of action can be detrimental to the principal. The transparency we consider is about the agent's innate ability.

¹⁷In Holmström (1999) a reduction in the dispersion of the distribution of ability reduces the effort of the agent because potential employers update less about ability when they observe output. This effect is absent in our model because we do not need to add an extra noise term. See also Alesina and Tabellini (2007) who use a career concerns model to contrast the outcomes for politicians and bureaucrats.

equilibrium policies. When transparency of ability is higher, then, this marginal increase in transfers has a bigger effect on the re-election probability.¹⁸ Hence, the incumbent's marginal benefit of such an increase is higher, leading to large transfers in equilibrium. Since transfers are partly financed through debt, debt also increases when transparency of ability is higher.

It is important to contrast this result with a result discussed in chapter 4 of Persson and Tabellini (2000). They consider a similar career concerns model with an incumbent government of uncertain ability that faces re-election against a challenger. In their model transparency of ability is beneficial because it leads to a reduction in corruption. When transparency of ability is higher, a marginal increase in corruption has a bigger effect on the re-election probability. Thus, the political cost of corruption is higher when transparency of ability is higher, leading to a lower level of corruption. The mechanism is very similar to the one highlighted in Proposition 3.3.

The contrasting effects of transparency in the two models can also be framed in terms of political accountability, if we think of the underlying mechanism at work in the two models. In both models the effects of increased transparency operate through a greater responsiveness of re-election probability to the incumbent's choices. We can interpret this responsiveness as a feature of the political system. In some political systems politicians are strongly accountable for their choices, in others less so. The view emerging from the analysis by Persson and Tabellini (2000) is that accountability is beneficial because it forces the politician to reduce corruption, or any activity that harms voters directly. In contrast, our result in Proposition 3.3 suggests that the effects of political accountability are more subtle in a world with imperfect transparency of transfers or debt. In such a world, politicians cannot be held fully accountable for intertemporal distortions. Thus, greater immediate accountability – i.e., greater responsiveness of re-election probability – may lead an incumbent to increase current wasteful spending at the cost of greater future taxes.

Because of the conflict between the welfare consequences of increased accountability in the two dimensions (transfers and corruption) it is not clear whether accountability is desirable. A possible way to resolve this conflict is to classify political systems into two broad categories according to the strength of their judicial institutions. In countries with weak institutions, there are few external constraints on corruption, so accountability is an important constraint on corruption. In countries with strong institutions, accountability is less likely to be the relevant constraint on corruption. Thus, in countries with strong institutions, the negative effects of accountability may dominate.

¹⁸This effect is similar to results in tournaments with handicaps where effort is higher the lower the handicap. See for instance Schotter and Weigelt (1992).

3.4 Inefficient Means of Redistribution and Manipulation of Public Accounts

In this Section, we discuss the incumbent's incentives to choose wasteful public spending, and to manipulate the public accounts—i.e., to “lie” to the public about the size of government. It is well known that governments transfer resources to special interests in an inefficient manner. For instance, there are large programs of farm subsidies—considered highly distortionary—although it would be more efficient to provide direct cash transfers or tax cuts. This has long been a puzzle in public economics, generating a debate about the causes of these inefficiencies.¹⁹ Furthermore, a common concern among international organizations such as World Bank, OECD and the IMF is the potential for abuse of public accounts. For example, in Hungary in September 2006, the prime minister Ferenc Gyurcsány accidentally revealed that his government had covered up the budget deficit's true size. Similarly, the Greek government manipulated the fiscal accounts from 2000 to 2003 in order to gain entry into the Eurozone. We now present a simple variation in the model of the previous Section that delivers both a choice of inefficient expenditures, as well as manipulation of the public accounts. It turns out that the efficiency consequences of such manipulations are more subtle than it may be expected from the simple discussions in the press.

Formally, we assume that the incumbent can transfer resources to each group via two alternative tools. The first tool is as described in the previous Section: when the incumbent chooses an amount of spending equal to y , voters receive $\theta = y + a$. The second tool is wasteful: when the incumbent chooses an amount of spending equal to z , voters receive $\theta = \lambda(z + a)$ where $\lambda \leq 1$ is a measure of the wastefulness of the z transfer. Moreover, for simplicity, we now assume that there is only one group (extending this analysis to n groups would be straightforward).

We first obtain a result describing the welfare consequences of the two alternative transfer tools. The next proposition compares two regimes: in the first regime the politician can only use the wasteful transfer tool; in the second regime the politician can only use the non-wasteful tool.

Proposition 3.4 *(i) For any value of α and any distribution $F(\cdot)$, there is a probability p and a wastefulness $\bar{\lambda}(p)$ such that if $\lambda \in [0, \bar{\lambda}]$, then social welfare is higher under the wasteful regime.*

(ii) If $\psi''' \leq 0$, then for all λ social welfare is higher under the wasteful regime.

Proof. See the Appendix ■

¹⁹For a theoretical model and a discussion of the literature, see Coate and Morris (1995).

The assumption that $\psi''' \leq 0$ is merely sufficient. In particular, the result holds strictly for a quadratic ψ .

It may seem surprising that welfare is higher when the government uses a wasteful transfer tool. Indeed, if government spending were fixed at a given level, then, of course, welfare would be lower in the wasteful regime. However, note that, even when transfers are not wasteful, government spending is inefficient because taxation is distortionary. When the transfer tool is wasteful and the level of government spending is chosen endogenously, the incumbent chooses to reduce spending relative to the level he would choose without waste. This reduces the distortion from taxation. Thus, there are two balancing forces. To gain some intuition for why the second effect dominates, consider the case of very wasteful transfers (i.e., very low λ). In this case, the government chooses to offer zero transfers because the marginal electoral gain from transfers is too low to compensate for the destruction of resources. The proof further shows that, when the marginal distortions from taxation are concave, the second effect dominates for any λ , so that welfare is higher under any level of waste. Indeed, the argument in the proof also shows that welfare is *increasing* in the degree of waste: the lower λ , the higher welfare.

It is interesting to consider a counterpart to the result in Proposition 3.4. If the government is currently operating in the wasteful regime, a reform banning the use of wasteful transfers would be harmful unless accompanied by some overall caps on spending. The reason is that the response by the government would be to increase spending so much that the distortions due to increased taxation would dominate the gains due to the elimination of wasteful transfers.

We now consider the incentives for the government to choose the wasteful transfer tool. We discuss two alternative scenarios: in the first scenario the politician can commit to the regime (wasteful or non-wasteful); in the second scenario the politician cannot commit.

Proposition 3.5 *If $\psi''' < 0$, and the incumbent can commit to the choice of transfer tool, then he chooses the wasteful transfer tool.*

Proof. By Proposition 3.4, if $\psi''' < 0$, welfare is higher in equilibrium under the wasteful regime. The ex-ante probability of re-election is 50 percent in both regimes because voters adjust their expectations to reflect the choice of transfer tool. Thus, the politician prefers the wasteful transfer tool. ■

It is useful to contrast this result with the analysis in Coate and Morris (1995). Coate and Morris provide a related explanation for the choice of inefficient means of redistribution. In their model, voters are uncertain about whether the incumbent is honest. If the incumbent chooses to transfer money to an interest group, then voters infer that the politician is dishonest

and he is voted out of office. If instead the incumbent chooses to build a public project that may benefit an interest group, then voters remain uncertain about the incumbent's honesty: a key assumption is that there is a positive probability that building the public project is beneficial. Coate and Morris show that, under these assumptions, politicians may choose to redistribute resources via wasteful public projects. There are two substantive differences between the result in Proposition 3.5 and the analysis in Coate and Morris. First, in our model the incumbent chooses the least efficient way of transferring resources even though its wastefulness is common knowledge. Second, in our model the incumbent *commits* to the choice of the inefficient transfer. Of course, both of these differences only take place because *in equilibrium* the choice of the more wasteful transfer tool leads to more efficient outcomes.

We now consider the scenario in which the politician cannot commit to the form of transfers.

It may seem unreasonable to assume that the incumbent can commit. We now show that, absent this commitment, the government would not choose the wasteful tool unless this tool offered some compensating advantage.

Proposition 3.6 *If the incumbent cannot commit to the choice of transfer tool, in equilibrium he chooses the non-wasteful transfer tool.*

Proof. Assume first that the incumbent chooses the non wasteful transfer tool and the size of transfers y^* are given by equation (5) (with $n = 1$). Suppose he deviates and chooses the wasteful tool, then, for any value of the wasteful transfers z , social welfare and re-election probability are lower than in the case in which the incumbent chooses the non-wasteful tool of the same amount—i.e., $y = z$. But y is dominated by y^* .

Assume now that the politician chooses the wasteful tool of some amount z^* . By choosing the non-wasteful tool of the same amount—i.e., $y = z^*$ —the incumbent raises both welfare and the re-election probability. ■

We now consider the possibility that the government has the opportunity to manipulate the public accounts in a way that may hide the true size of transfers. Specifically, we allow the possibility that the wasteful transfer lends itself to such manipulation. In order to capture the phenomenon of the abuse of public accounts, we assume that the wasteful tool is amenable to manipulation: in the event that voters observe total spending, instead of observing z , they observe πz , with $\pi < 1$. We assume that the probability p of observing total spending is the same for both tools. This is a useful assumption because it allows us to isolate the effect of manipulation.

The next proposition shows that the wasteful transfer is likely to be chosen when: the degree of wastefulness is not too high (high λ), when the level of transparency is relatively high (high p), and when the manipulability is high (low π).

Proposition 3.7 *For any $0 < p < 1$, and $\pi < 1$, there exists a wastefulness $\lambda(p, \pi) < 1$ such that if $\lambda(p, \pi) < \lambda \leq 1$, in equilibrium the incumbent chooses the wasteful transfer. Furthermore, $\lambda(p, \pi)$ is decreasing in p and increasing in π .*

Proof. See Appendix. ■

The intuition for the result of Proposition 3.7 is that, if λ is high enough, then the incumbent obtains only a small increase in his re-election probability by choosing the non-wasteful transfer. Since social welfare is higher when using the wasteful transfer, then the incumbent prefers to use the wasteful transfer. Moreover, when p is very close to 1, voters are very likely to learn the incumbent's ability, so the incumbent's incentives to affect the re-election probability are lower. Hence, he chooses the wasteful transfer since it maximizes social welfare.

Taken together, the analysis in this Section suggests that forbidding the manipulation of public accounts may not be a panacea because it may induce an increase in the amount of spending. This may seem paradoxical, and is clearly counter to common wisdom. Our results can be reconciled with the standard view of such manipulations, if we assume that voters are in fact naive about such accounts. Proposition 3.7 relies crucially on voters' sophistication: in an equilibrium with manipulation, voters understand that when they observe a given level of spending, the true amount spent must be inflated by $\frac{1}{\pi}$ to adjust for the fiscal manipulation. If instead voters took at face value the amount of spending reported, then it can be shown that the level of distortions chosen by the incumbent government would increase.²⁰

3.5 Competitiveness of the Political System

Thus far we have assumed that the challenger and the incumbent are ex-ante identical, i.e., they have abilities that are drawn from the same distribution. We now consider what happens if the challenger is “weaker” than the incumbent (or vice-versa). We model this by assuming that the challenger's abilities have lower mean than the incumbent's. Namely, we assume that the incumbent's ability has mean 0 and the challenger's ability has mean $-c < 0$. We also assume that the incumbent's ability has density f that is symmetric and single peaked around

²⁰Milesi-Ferretti (2003) studies the effects of fiscal rules when the government has a margin for “creative accounting.” He shows that the effects of increased transparency on the cyclical response of the budget to shocks are ambiguous.

its mean. We can think of the degree of inferiority of the challenger as describing the average competitiveness of the race.

Proposition 3.8 *The size of government is increasing in the degree of competitiveness of the political system.*

Proof. Here we only prove the result for the case where ability on taxes is degenerate but it is easy to show that the Proposition extends to the more general case. The equilibrium level of spending $y^*(c)$ is given by

$$\psi'(y^*(c)) = 1 + \frac{\alpha(1-p)}{1-\alpha} f(c).$$

Since ψ is convex and f is single-peaked at zero, we obtain that $y^*(c)$ is increasing in c . ■

The intuition for this result is the following. In this model the incentive to offer wasteful transfers is given by the incumbent's re-election concerns. If the challenger presents a lower threat (because of lower ability), the incumbent's incentive to manipulate the electoral process are reduced.²¹

Our measure of competitiveness is consistent with the one used by Alt and Lassen (2006). They use a measure of turnover as an instrument for their transparency variable. One of their tables shows a negative relation between turnover and debt. This is in contrast with Proposition 3.8. One possible way to reconcile our theoretical result with the correlation in Alt and Lassen (2006) is to allow for a reason for turnover that is unrelated to the difference in abilities among competing politicians. To model this, we assume an exogenous probability ζ of government termination. As ζ increases, our incumbent politician would endogenously place more weight on social welfare, and, as a result, reduce debt. Thus, this simple extension can reconcile the result of our model with the correlation in Alt and Lassen (2006).

²¹There is a growing literature that addresses the effects of electoral competition on government policy. Thus far, the literature has presented two distinct notions of electoral competition. Lizzeri and Persico (2005) define competition as the number of competing candidates. They show that increasing the number of competing candidates lowers welfare by shifting spending toward targetable transfers and away from beneficial public goods. Polo (1998) shows that candidates' uncertainty about the position of the median voter can relax political competition, thereby allowing candidates to retain rents. Rogers and Rogers (2000) look at data from US gubernatorial elections and find some evidence that closer elections are associated with tighter budgets. Besley et al. (2007) present a probabilistic voting model in which increasing the dominance of one party leads to the selection of candidates of inferior ability, increased corruption, and lower economic performance. They present evidence from the US South that is consistent with the model.

Appendix: Omitted Proofs

Proof of Lemma 3.1. To simplify notation, let

$$\beta_i = 1 - F(y_i^* - y_i)$$

be the probability that group i vote for the incumbent.

In event (i), the incumbent is re-elected if at least $\frac{n+1}{2}$ groups vote for him. Hence, the probability that the incumbent receives the vote of at least $\frac{n+1}{2}$ groups is equal to:

$$\begin{aligned} & \beta_1 \beta_2 \dots \beta_n + \\ & + \beta_1 \beta_2 \dots \beta_{n-1} (1 - \beta_n) + \dots + (1 - \beta_1) \beta_2 \dots \beta_n + \\ & \dots + \\ & + \beta_1 \dots \beta_{\frac{n+1}{2}} \left(1 - \beta_{\frac{n+1}{2}+1}\right) \dots (1 - \beta_n) + \dots + (1 - \beta_1) \dots \left(1 - \beta_{\frac{n-1}{2}}\right) \beta_{\frac{n+1}{2}} \dots \beta_n. \end{aligned}$$

The above expression can be rewritten as

$$\begin{aligned} & \beta_1 (\beta_2 \dots \beta_n - \beta_2 \dots \beta_n) + \beta_1 (\beta_2 \dots \beta_{n-1} (1 - \beta_n) - \beta_2 \dots \beta_{n-1} (1 - \beta_n)) + \dots \\ & \dots + \\ & \beta_1 \left((1 - \beta_2) \dots \left(1 - \beta_{\frac{n-1}{2}}\right) \beta_{\frac{n+1}{2}} \dots \beta_n - (1 - \beta_2) \dots \left(1 - \beta_{\frac{n-1}{2}}\right) \beta_{\frac{n+1}{2}} \dots \beta_n \right) + \\ & + \beta_1 \left(\beta_2 \dots \beta_{\frac{n+1}{2}} \left(1 - \beta_{\frac{n+1}{2}+1}\right) \dots (1 - \beta_n) + \dots + (1 - \beta_2) \dots \left(1 - \beta_{\frac{n+1}{2}}\right) \beta_{\frac{n+1}{2}+1} \dots \beta_n \right) \\ & + \beta_2 \dots \beta_n + \beta_2 \dots \beta_{n-1} (1 - \beta_n) + \dots + (1 - \beta_2) \dots \left(1 - \beta_{\frac{n-1}{2}}\right) \beta_{\frac{n+1}{2}} \dots \beta_n \end{aligned}$$

Cancelling terms, the above expression simplifies to

$$\begin{aligned} & \beta_1 \left(\beta_2 \dots \beta_{\frac{n+1}{2}} \left(1 - \beta_{\frac{n+1}{2}+1}\right) \dots (1 - \beta_n) + \dots + (1 - \beta_2) \dots \left(1 - \beta_{\frac{n+1}{2}}\right) \beta_{\frac{n+1}{2}+1} \dots \beta_n \right) \\ & + \beta_2 \dots \beta_n + \beta_2 \dots \beta_{n-1} (1 - \beta_n) + \dots + (1 - \beta_2) \dots \left(1 - \beta_{\frac{n-1}{2}}\right) \beta_{\frac{n+1}{2}} \dots \beta_n \end{aligned}$$

which can be expressed as

$$\beta_1 k(\beta_{-1}) + m(\beta_{-1})$$

where $k(\beta_{-1})$ and $m(\beta_{-1})$ are functions of all probabilities β_i for $i \neq 1$.

Moreover, for any equilibrium strategy profile $(y_1^*, \dots, y_n^*)$, $(t_1^*, \dots, t_n^*)$ and deviation (y_1, y_{-1}^*) , (t_1, t_{-1}^*) we have that

$$y_i = y_i^* \text{ and } s(t_i) = s(t_i^*) \text{ for all } i \neq 1$$

Therefore

$$\beta_i = 1 - F(0) = 1/2 \text{ for all } i \neq 1.$$

In addition, $\beta_1 k(\beta_{-1})$ is the probability of the event in which group 1 is pivotal for the re-election, i.e. $\frac{n-1}{2}$ of the other $n-1$ groups vote for the incumbent. Hence

$$\begin{aligned} k(\beta_{-1}) &= \left(\beta_2 \dots \beta_{\frac{n+1}{2}} \left(1 - \beta_{\frac{n+1}{2}+1} \right) \dots (1 - \beta_n) + \dots + (1 - \beta_2) \dots \left(1 - \beta_{\frac{n+1}{2}} \right) \beta_{\frac{n+1}{2}+1} \dots \beta_n \right) \\ &= \binom{n-1}{\frac{n-1}{2}} \left(\frac{1}{2} \right)^{n-1} \end{aligned}$$

as we are choosing exactly $\frac{n-1}{2}$ groups (voting in favor of the incumbent) out of $n-1$ groups (group 1 is excluded).

The analysis obviously generalizes to any group i . The lemma thus follows. ■

Proof of Proposition 3.4. First, note that in the non wasteful regime Social Welfare is given by $SW(y^*) = y^* - \psi(y^*)$, while in the wasteful regime it is given by $SW(z^*) = \lambda z^* - \psi(z^*)$, where y^* and z^* are the equilibrium choices under the non wasteful and wasteful regimes, respectively.

In order to prove the results, we first need to characterize y^* and z^* , i.e. the equilibrium choices under the two regimes. Assume first that voters expect the incumbent to choose y . This is a special case of the analysis above and we obtain that the first order condition is

$$\psi'(y^*) = 1 + \frac{\alpha(1-p)}{1-\alpha} f(0). \quad (7)$$

Assume now that voters expect the incumbent to choose z . In the event that voters observe nothing, consider a voter who observes g . If he conjectures $\hat{z}$, then

$$g = \lambda(\hat{z} + \hat{a}) \text{ so } \hat{a} = \frac{g}{\lambda} - \hat{z} = z + a - \hat{z}$$

implying that candidate wins if $a > 0$, i.e., $z + a - \hat{z} > 0$, i.e., $a > \hat{z} - z$ so, probability of victory is given by

$$1 - F(\hat{z} - z).$$

In the event that voters observe z , then probability of winning is $1/2$.

Since social welfare is given by $\lambda z - s(z)$, we obtain that the candidate seeks to maximize

$$\alpha \left((1-p)(1 - F(\hat{z} - z)) + p \frac{1}{2} \right) + (1-\alpha)(\lambda z - \psi(z)).$$

The first order conditions for interior solutions are given

$$\alpha(1-p)f(0) = (1-\alpha)(\psi'(z) - \lambda).$$

Rearranging this equation and taking into account the fact that, for some parameter values, the incumbent chooses the corner solution of $z = 0$, we obtain the characterization:

$$\psi'(z^*) = \max \left\{ 1, \lambda + \frac{\alpha(1-p)}{1-\alpha} f(0) \right\}. \quad (8)$$

This implies that, under the wasteful regime, if $\lambda < 1$, then for p sufficiently high, the incumbent chooses $z^* = 0$ whereas, for any $p < 1$, in the no waste regime, $y > 0$. Thus, there is always a set of parameters for which welfare is higher under the wasteful regime. This proves part (i).

As for part (ii), we now compare $SW(y^*)$ and $SW(z^*)$. First notice that welfare is clearly the same in the two regimes if $\lambda = 1$. Moreover, if $\lambda = 0$, then obviously there is no electoral gain in offering transfer z but only costs in terms of welfare; the incumbent thus chooses $z^* = 0$. Similarly, as long as

$$\lambda + \frac{\alpha(1-p)}{1-\alpha}f(0) \leq 1$$

the incumbent chooses $z^* = 0$.

Thus, consider any interior λ , i.e. $\lambda \in \left[1 - \frac{\alpha(1-p)}{1-\alpha}f(0), 1\right]$. In that range, the difference in welfare is equal to

$$SW(y^*) - SW(z^*) = y^* - \psi(y^*) - \lambda z^* + \psi(z^*)$$

where z^* solves $\psi'(z^*) = \lambda + \frac{\alpha(1-p)}{1-\alpha}f(0)$. We now that that $SW(y^*) - SW(z^*)$ is increasing in λ . Since at $\lambda = 1$, the difference is equal to 0, this implies that for all interior λ strictly less than 1, social welfare is higher under the fraudulent regime.

Taking the derivative with respect to lambda of the difference $y^* - \lambda z^* - \psi(y^*) + \psi(z^*)$, we obtain

$$-z^* - \lambda \frac{\partial z^*}{\partial \lambda} + \frac{\partial s}{\partial z^*} \frac{\partial z^*}{\partial \lambda} = -z^* + (\psi'(z^*) - \lambda) \frac{\partial z^*}{\partial \lambda} \quad (9)$$

From the equilibrium condition $\psi'(z^*) = \lambda + \frac{\alpha(1-p)}{1-\alpha}f(0)$, we obtain that

$$\psi'(z^*) - \lambda = \frac{\alpha(1-p)}{1-\alpha}f(0) \text{ and } \frac{\partial z^*}{\partial \lambda} = \frac{1}{\psi''(z^*)}$$

Thus, equation (9) is equal to

$$-z^* + \frac{\alpha(1-p)}{1-\alpha}f(0) \frac{1}{\psi''(z^*)} \quad (10)$$

We want to obtain the sign of equation (10). We can multiply equation (10) by $\psi''(z^*) > 0$, and the sign of equation (10) is than the same as the sign of

$$-\psi''(z^*) z^* + \frac{\alpha(1-p)}{1-\alpha}f(0). \quad (11)$$

Using a Taylor expansion, we know that

$$\psi'(0) = \psi'(z^*) - \psi''(z^*) z^* + \frac{\psi'''(k) z^{*2}}{2} \quad (12)$$

for some k between 0 and z^* . Thus, rearranging equation (12), we obtain

$$\psi''(z^*) z^* = \psi'(z^*) - \psi'(0) + \frac{\psi'''(k) z^{*2}}{2}.$$

Substituting the expression for $\psi'(z^*) = \lambda + \frac{\alpha(1-p)}{1-\alpha} f(0)$ and $\psi'(0) = 1$, we obtain

$$\psi''(z^*) z^* = \lambda + \frac{\alpha(1-p)}{1-\alpha} f(0) - 1 + \frac{\psi'''(k) z^{*2}}{2} \quad (13)$$

Substituting equation (13) in equation (11)

$$\begin{aligned} -\psi''(z^*) z^* + \frac{\alpha(1-p)}{1-\alpha} f(0) &= -\left(\lambda + \frac{\alpha(1-p)}{1-\alpha} f(0) - 1 + \frac{\psi'''(k) z^{*2}}{2}\right) + \frac{\alpha(1-p)}{1-\alpha} f(0) \\ &= -\left(\lambda - 1 + \frac{\psi'''(k) z^{*2}}{2}\right) > 0 \end{aligned}$$

since $\lambda - 1 < 0$ and $\psi'''(k) < 0$. ■

Proof of Proposition 3.7. We characterize the conditions under which an equilibrium with wasteful transfers exists. Similar calculations show that the same conditions necessitate such an equilibrium.

Assume voters believe that the incumbent chooses the wasteful transfer. Then, the equilibrium z^* is given by equation (8). Consider a deviation in which the incumbent chooses the non-wasteful transfer with a level y . Then, there are two events. In the event that voters observe θ , we have

$$\theta = y + a$$

but voters believe that $\theta = \lambda(z^* + \hat{a})$. Thus, $\hat{a} = \frac{\theta}{\lambda} - z^* = \frac{y+a}{\lambda} - z^*$, implying that the probability of re-election is given by the probability that $\frac{y+a}{\lambda} - z^* > 0$, or that $a > \lambda z^* - y$. Thus, probability of re-election in this event is

$$1 - F(\lambda z^* - y)$$

In the event that voters observe y , they believe that, in fact, spending is $\frac{y}{\pi}$. Thus, they believe that $\hat{a} = \theta - \frac{y}{\pi} = y + a - \frac{y}{\pi}$. Thus, the probability of re-election in this event is

$$1 - F\left(y \frac{1-\pi}{\pi}\right).$$

Since welfare is given by $y - s(y)$, we obtain that the incumbent's payoff from this deviation is

$$\alpha \left[(1-p)(1 - F(\lambda z^* - y)) + p \left(1 - F\left(y \frac{1-\pi}{\pi}\right) \right) \right] + (1-\alpha)(y - \psi(y)).$$

This implies that the difference in payoffs relative to the candidate equilibrium is given by

$$\alpha \left[(1-p)(1-F(\lambda z^* - y)) + p \left(1 - F \left(y \frac{1-\pi}{\pi} \right) \right) - \frac{1}{2} \right] + (1-\alpha)(y - \psi(y) - (\lambda z^* - \psi(z^*))) .$$

We want to find the set of parameters such that, the maximized value of the above expression with respect to y , is negative. The expression is decreasing in λ and increasing in π . Furthermore, it is easy to verify that, if $y > \lambda z^*$, then the expression is decreasing in p . Since the maximizing y satisfies the inequality, the result follows. ■

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